

From HVAC to Home Performance Contracting

Matthew Holtkamp knew when he was in technical school at Indian Hills Community College in Ottumwa, Iowa, that he would have his own HVAC business some day. In 1996 his dream became a reality when he founded Holtkamp Heating and Air Conditioning (HHVAC)—a company specializing in service, preventive maintenance, and replacement of existing HVAC systems—in Atlanta, Georgia. With Holtkamp's expertise and diligence and his wife Suzanne's marketing and business skills, the company had what it takes to succeed. Holtkamp is a charismatic and outgoing guy, so HHVAC got a lot of referrals from friends and acquaintances. And by focusing on existing homes, Holtkamp increased his new company's chance of success. Several of his friends had begun businesses that supplied HVAC services to the new-construction market—and had failed because margins for new-homes installations were much tighter than margins for existing-homes installations.

HHVAC was profitable from year one. Holtkamp hired a third employee, a service technician, in 1997. Once again, his business exceeded expectations, and sales increased from \$150,000 in the first year to \$250,000 in the second year. Revenue kept growing as Holtkamp hired more staff. In 2002 the company reached a plateau with one installer and two service technicians, and HHVAC averaged \$800,000 in business per year from 1999 to 2002.

Callbacks Are a Killer

Financially the company was doing very well, but there was one problem. Three times in six months clients had complained that "the new unit didn't save me any money, and it says right here in the brochure, 'SAVE 50% on your cooling bill.'" What was even worse was that in two of these cases, the new high-efficiency equipment was providing less cooling than the old buildergrade



MATTHEW HOLT KAMP

Jeff Huddleston of Holtkamp Heating and Air Conditioning finishes up a home energy audit.

unit. Holtkamp had to find a solution to this embarrassing problem—the good reputation and the future success of his company depended on it.

During the visits to the customers that had complained about inadequate cooling, the technicians investigated the ducts and found an unusually large amount of air escaping from the duct connections. Although Holtkamp knew from experience to examine the ducts for leakiness, he didn't yet understand the dynamics of how the whole house was performing and was unsure of the effects of installing new HVAC equipment. The source of the inefficiency in the troubled HVAC systems was air leakage from the ducts, poor insulation, and leaky building envelopes. The ducts leaked in part because the HVAC equipment was improperly installed. During the installation, HHVAC personnel sometimes accidentally loosened the ductwork. Then, when the customer complained about the new equipment's poor performance, the company's solution was to install a new plenum. During that installation, the ductwork was

loosened up even more, resulting in even more air leakage.

The source of the problem was identified. HHVAC's future success depended on finding out how to prevent the problem from happening again.

Training Makes the Difference

The answer turned out to be one of training. One day, while he was mulling over his company's callback problem, Holtkamp received an ad for an informational building science class offered by The Comfort Institute. "This knowledge will allow you to specialize," the ad said; "—you can become an indoor air specialist." Something else in the ad caught Holtkamp's attention. "Why is it that our air filter works in one home and not in the next?" said a quote attributed to the air filter company Aprilaire. Holtkamp realized that this exactly described his own experience with Aprilaire filters. Some of his customers said that the filters were great, while others said that they were a waste of money. A whole-house approach to HVAC installation could make the

difference between an efficient, comfortable home and an inefficient and uncomfortable one.

Holtkamp was curious to see what The Comfort Institute had to offer and whether this training might help him find a solution to his problem, so he immediately signed up himself and the staff for a one-day class offered in Atlanta by The Comfort Institute's Ken Summers. The principles of building science were taught in the classroom and demonstrated with a Plexiglas model home that simulates what occurs with different air pressures in the home.

Out of the 30 contractors attending the one-day class, only one signed up for the full training—and that one was Holtkamp. It was definitely worth it. All the diagnostic and remediation procedures were clearly explained and illustrated inside the demonstration house. Holtkamp was particularly impressed by the blower door test, so impressed that he decided to use it on every estimate from then on. And the benefits of the course didn't end there. The Comfort Institute people offered to go with Holtkamp to his worst-case customer's home to diagnose the home's seemingly unsolvable problems. The house was uncomfortably damp, had mold growing around the registers, and the energy bills were outrageous—the house was a breeding ground for callbacks. They used a blower door and other specialized equipment to identify the source of these problems and to recommend fixes. It turned out that the insulation was in pathetic shape, the house had an excessively leaky shell, leaky ducts, and the newly installed heating and A/C equipment was oversized.

The Comfort Institute comprehensive building science training turned out to be 3 times more expensive than Holtkamp had expected—around \$20,000, including equipment—but it was worth every penny. Holtkamp and all of his employees at the time attended the full training. Holtkamp's company was also qualified to participate in the Home Performance with Energy Star program, which requires the kind of whole house approach to diagnostics

and testing that The Comfort Institute taught him and his technicians.

Before receiving training, if a client had a dust problem, Holtkamp would recommend a high-efficiency filter. He knew that the filter may not solve the problem, but he had no other solution to offer, and no way to be 100% sure of the outcome. With the new testing equipment and procedures, Holtkamp could pin down the source of any HVAC problem and find a solution based on the condition of the whole house. Before the training, for example, if the client complained of high energy bills and dust, he would install a new high-efficiency furnace and recommend a new furnace filter for dust. After the training, he and his employees would diagnose the problem as duct leakage using a blower door assessment, and seal the ducts before installing the new furnace. When Holtkamp hires new employees, he immediately schedules them for The Comfort Institute training.

After The Comfort Institute training, Holtkamp can guarantee to his clients that they are getting what they are paying for. At the same time, he can prove that a 14-SEER unit installed by a competitor who did not take the whole-house approach would give them the equivalent of 11 SEER or less.

Sweet Smell of Success (and Good IAQ)

Holtkamp's investment paid for itself in less than a year. His company took the training in January 2003. By March 2003 revenues were increasing exponentially, the company's margins had tripled, and the company was experiencing higher closing rates than ever before. The company's average job had increased from \$6,500 for a new A/C and furnace installation to well over \$10,000, including air sealing and duct sealing. Referrals too increased considerably, as satisfied homeowners became strong advocates of the whole-house approach; they now constitute more than 80% of Holtkamp's new business. Finally, the new approach helped Holtkamp to distinguish his business from that of his competitors.

HHVAC gets at least one call a week from happy customers. One customer was so astonished by the results of HHVAC's work on his HVAC system that the next day he called to say, "For the first time I woke up and I could breathe." Success cases like this one motivate Holtkamp and the staff of HHVAC to keep on doing the best job they can to turn each customer into a healthier and happier customer.



—Patricia Plympton

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Home Performance with Energy Star, a program sponsored nationally by DOE and EPA, and available in select cities, offers a comprehensive, whole-house approach to making energy-efficient home improvements. The program is managed locally by a program sponsor (typically a utility company, state agency, or local association), which recruits and trains contractors and helps ensure they deliver quality work. The local sponsor may also offer financial assistance. Participating contractors provide recommendations and services to homeowners to make their homes more comfortable and energy efficient, while reducing their utility bills and helping to protect the environment.

For more information on Home Performance with Energy Star, visit www.energystar.gov/homeperformance, or call (888) STAR-YES ((888)782-7937).